

EMAIL HEADER ANALYSIS & PHISHING INVESTIGATION

SOC Lab · Threat Intelligence Report · December 2025

■ VERDICT: SUSPICIOUS / BEC SCAM EMAIL

Project	Email Header Analysis & Threat Intelligence Investigation (SOC Lab)
Objective	Determine whether a suspicious email is malicious or benign using header analysis, WHOIS, IP reputation, and threat intelligence correlation.
Sender	Dr. Franklin Reece <pbs17192@pbs.ac.th>
Subject	"Good Morning,"
Received	Tue, 16 Dec 2025 · Yahoo Mail · Spam Folder

SKILLS DEMONSTRATED

■ Email Header Analysis	■ SPF / DKIM / DMARC Validation
■ WHOIS Domain Analysis	■ IP Reputation Analysis
■ Threat Intelligence Correlation	■ SOC Investigation & Documentation

01

SUSPICIOUS EMAIL IDENTIFICATION

A suspicious email titled **"Good Morning,"** was received in the **Spam folder** of Yahoo Mail. The sender claimed to be a foreign supplier requesting business engagement — a classic **Business Email Compromise (BEC)** tactic.

OBSERVED RED FLAGS

- ✗ Generic subject line — **"Good Morning,"**
- ✗ Business solicitation from unknown, unverified foreign sender
- ✗ Email delivered directly to the **Spam folder**
- ✗ External sender domain (**pbs.ac.th**) — Thai academic institution
- ✗ **Reply-To** address differed from the sender domain

SCREENSHOT — Yahoo Mail Spam Folder

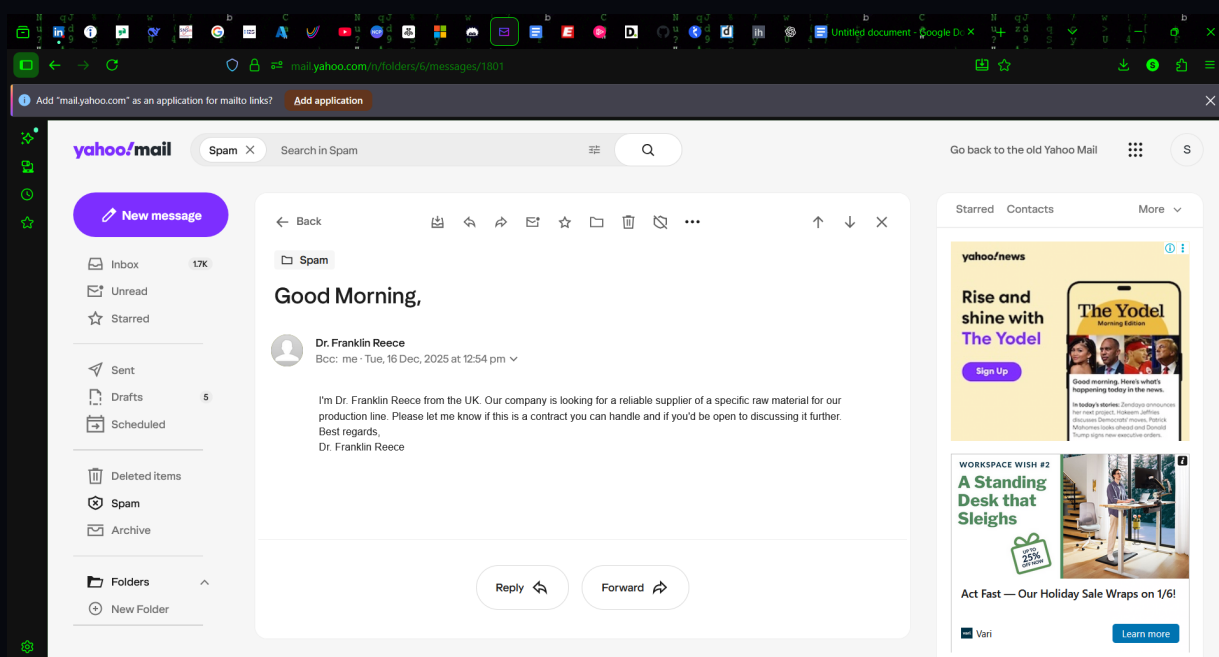


Fig 1. Suspicious email visible in Yahoo Mail spam folder.

02 EMAIL HEADER EXTRACTION

Since Yahoo Mail does not provide a direct "Copy Headers" option, the **View Raw Message** feature was used to access the complete SMTP header chain.

1. Opened the suspicious email in Yahoo Mail
2. Selected **More options** → **View Raw Message**
3. Copied the entire raw email header for analysis

SCREENSHOT — Raw Email Header

```
Received: from 10.197.37.232
by atlas107.free.mail.bf1.yahoo.com pod-id NONE with HTTPS; Tue, 16 Dec 2025 07:24:20 +0000
Return-Path: <pbs17192@pbs.ac.th>
X-Originating-IP: [209.85.219.66]
Received-SPF: none (domain of pbs.ac.th does not designate permitted sender hosts)
Authentication-Results: atlas107.free.mail.bf1.yahoo.com;
dkim-pass header.i@pbs.ac.th header.s=google arc_overridden_status=NOT_OVERRIDDEN;
spf=none smtp.mailfrom=pbs.ac.th arc_overridden_status=NOT_OVERRIDDEN;
dmarc-unknown header.from=pbs.ac.th arc_overridden_status=NOT_OVERRIDDEN;
X-Apparently-To: shaiivakumar16@yahoo.com; Tue, 16 Dec 2025 07:24:20 +0000
X-MailSize: M.Buc9cWLDuZL8jTpn48U7Kzbb.YuFINAu21849YchNATG61
70rqnetSv1BQlM4jpyW07B8pIRK3nk.0dHCpYpGY7d46kJG8BR_mKFNZXU1S
Zg62a1kglUzKQ11v8nB0n5x94R.WhiVva0225c4LExMoNq9GQIPs6vIj1LUK
1LSi5dFFmK2u5jUG19c35vBY1NCpZQKndecxZgsh9pXt8Kocf.10dFT
R2N_A0M1rb1hpy76Pw6saanaAQcPTRVX5x6ACOGU_wjGUUh8NlIp.mqR_K
Mzpnv09o9yqZj2A7yodndwQvY9Np89vvrzF52Jv8Ref_RekB8gRrWwvRuNI
1t21Vh0gupGhXjTRXZYh1ez152PVEZDLm6dylV6_ZZuZ9V5_GCSgr4r2gK
LDAV7OLLtowlL68Gyo_Q0fthKcFts11hVvYwWj3jYVha8eT.onixadu7HB
fNx4ZGYQg683wBCY5mcjy1D1DbLbWv7z4P_hEGXJ26Y1B31Mw3y5xkX
m7MIshvREgk04.o9hZRMvXv8xkbVJqoZ8IXEAFxUMpocX78bZQ1h_mtqp5CO
rXAdylZcKk7FV9b:johinhhKnZY40X.pXV2WEaQ59ez9g_cNEuXB1vrw64p_
EHednz7HRZIEE_mCTC1CwMvrr60KfL3.7bU0lyc93J0y3o3y9ShmDn.Oa
bW91bFrQWQXa7f31vX0D4VvY1nTW9V7d3mb5XRL1v_p6fkyYfW64kDue
HRoz15nXKQinb3J4U56VhEjRlmp2CznU25JXF_mxnOr1L2aTcz2_3r3nz
Ietq3rCluPkiW4dAALPs1hCghWjSY0XospX5VPx1GLNzo17QintBy0M4rY9
vPSLxQJ0uM.22eQ2vDm4DvU3uRbV48XK8919Q2gBnXp6C_lyfHf8TWKIT
CRRV823xy2ZhdHf9qPwDOrT8o5k833ZRBfXByoneIm0wQfHEp7g1Cpik
Udc6BGQ1or8s789uu9BTKAnYecb5QPvwm1LgyoJGVVUKP8U055gVwv12FeAA
3013JK17Mq1BQ4p2HhAKDks120F32N1jmc1L4c3a1Vr-ICPrVXqV0KjP0F2
_p1C1B00MPz126a4t35pC3ai3yOXi4da7dDqZelFDB1IKfHF-6tveb
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7MaBoxumW0VT5J0612L2F0645B51.wcX8RsnR4kwSD
Received: from 209.85.219.66 (EHL0 mail-qv1-f66.google.com)
by 10.197.37.232 with SMTPs
(version=TLS1.3 cipher=TLS_AES_128_GCM_SHA256);
Tue, 16 Dec 2025 07:24:20 +0000
Received: by mail-qv1-f66.google.com with SMTP id 6a1803df08f44-88a288811a4so26854486d6.3
for <shaiivakumar16@yahoo.com>; Mon, 15 Dec 2025 23:24:20 -0800 (PST)
DKIM-Signature: v=1; a=rsa-sha256; c=relaxed/relaxed;
d=pbs.ac.th; s=google; t=1765869859; x=1766474659; darn=yahoo.com;
h=to:subject:message-id:date:from:reply-to:mime-version:from:to:cc
:subject:date:message-id:reply-to;
bh=0tr1EX9MEks5G1t1uYQ2AvczPFDLhdsxln3HGIVPtc=;
b=C3e0Bef//d/kH:6Fsp1oiHh7ARSu00P7kRBelpOtBo71/IyKxRow7/4psA/cM+mJg
s1/vs2l13tph5N07H3IH+EOgg54rTJUtTz+2g/+6VwGkrBwBvCL1YrUqHqV3h4Qq
QYcMy4dwz2tqd1G1Ay5aZcPGBH1U3B9QwDpa7ZKEGYFHYq9w55QpVaF1S85x54AcS
vqt19/11uK1pFmTeKvY773x8e07H1sow710H8Pc4u5y/nfF8p8E5XK1Vg36c82J0Hx
```

Fig 2. Raw email header viewed via Yahoo Mail View Raw Message.

03

HEADER ANALYSIS (Google Admin Toolbox)

The raw headers were pasted into the [Google Admin Toolbox – Messageheader Analyzer](#) for structured parsing and hop-by-hop delivery inspection.

Message Origin	Google mail servers (mail-qv1-f66.google.com)
Sender Domain	pbs.ac.th
Relay Path	Routed through Gmail infrastructure
Delivery Time	Normal — no anomalous delays detected
Originating IP	209.85.219.66 (Google LLC / AS15169)

SCREENSHOT — Google Admin Toolbox Messageheader Analyzer

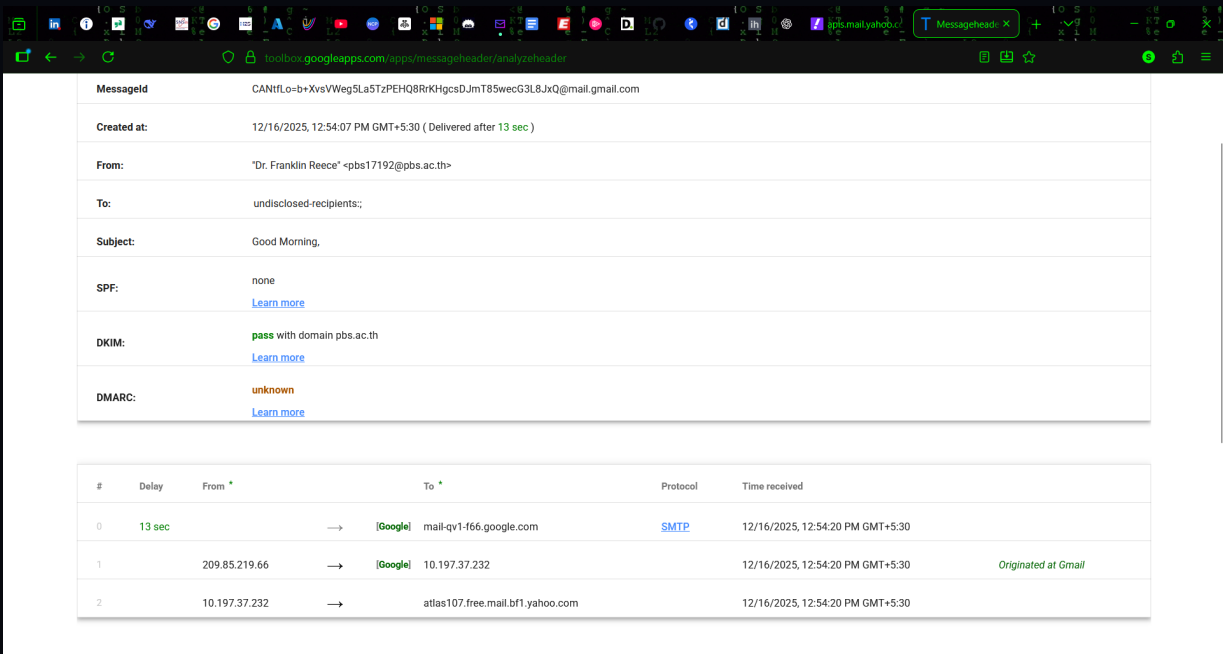


Fig 3. Header analysis results in Google Admin Messageheader Analyzer.

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AUTHENTICATION RESULTS (SPF / DKIM / DMARC)

MECHANIS M	RESULT	OBSERVATION	RISK LEVEL
SPF	NONE	Domain did not explicitly authorise the sender	MEDIUM
DKIM	PASS	Email cryptographically signed — integrity confirmed	LOW
DMARC	UNKNOWN	No DMARC policy — domain unprotected against spoofing	HIGH

■ **Security Insight:** Even though DKIM passed, **SPF = none** and **DMARC = unknown** reduce trust and indicate a **weak email security posture** for the sending domain.

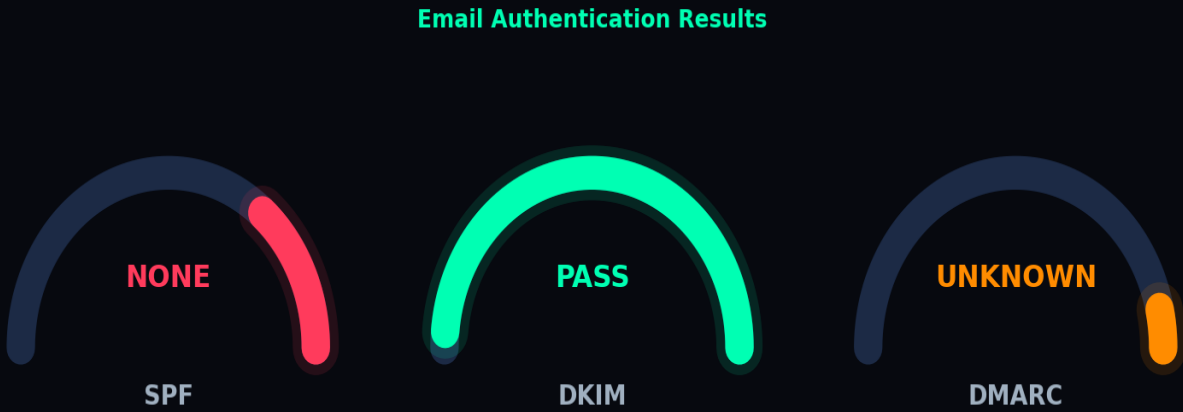


Fig 4. SPF / DKIM / DMARC authentication status gauges.

05

SENDER DOMAIN WHOIS ANALYSIS

The sender domain **pbs.ac.th** was investigated via WHOIS lookup to verify registration details and assess sender legitimacy.

Domain	pbs.ac.th
Registered	2002-12-12 (23+ years active)
Expires	2026-12-11
Last Updated	2025-11-19
Registrar	T.H.NIC Co., Ltd — IANA ID: 10001
Status	Active
Name Servers	haley.ns.cloudflare.com / kayden.ns.cloudflare.com
DNSSEC	Unsigned

■ **Conclusion:** Domain appears legitimate and long-standing (registered 2002). However this does **not** guarantee sender legitimacy — accounts on legitimate domains can be compromised or spoofed.

SCREENSHOT — WHOIS Lookup (GoDaddy)

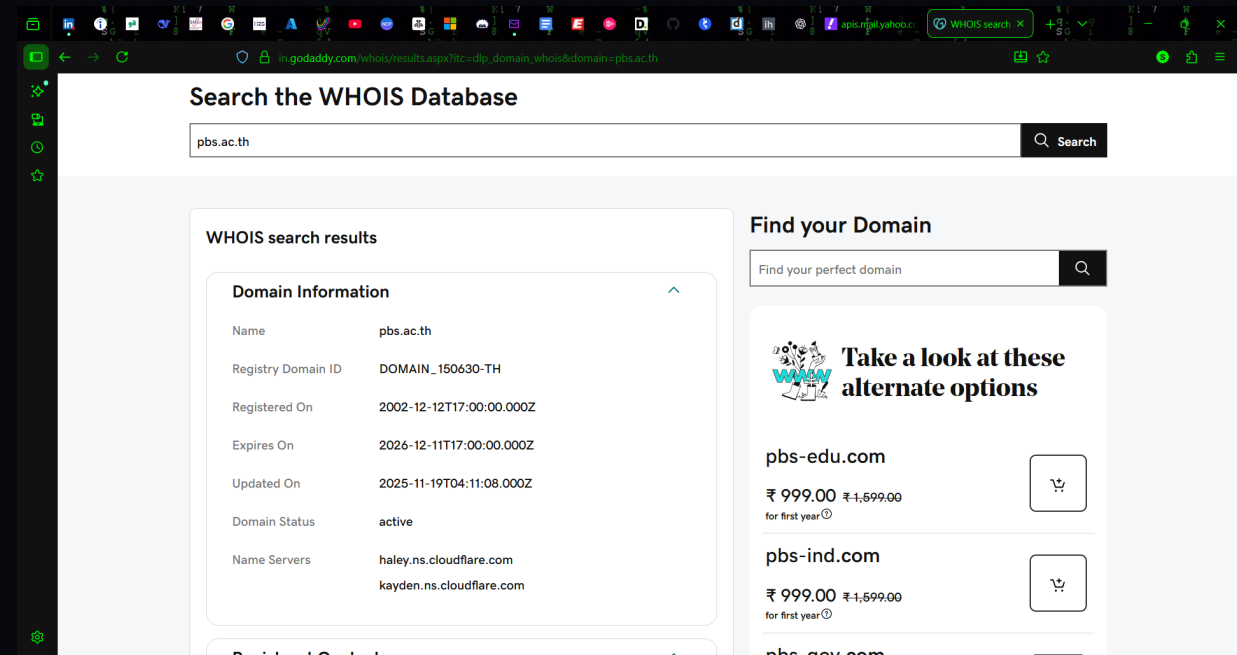


Fig 5. WHOIS lookup results for pbs.ac.th.

SCREENSHOT — WHOIS Registrar Details

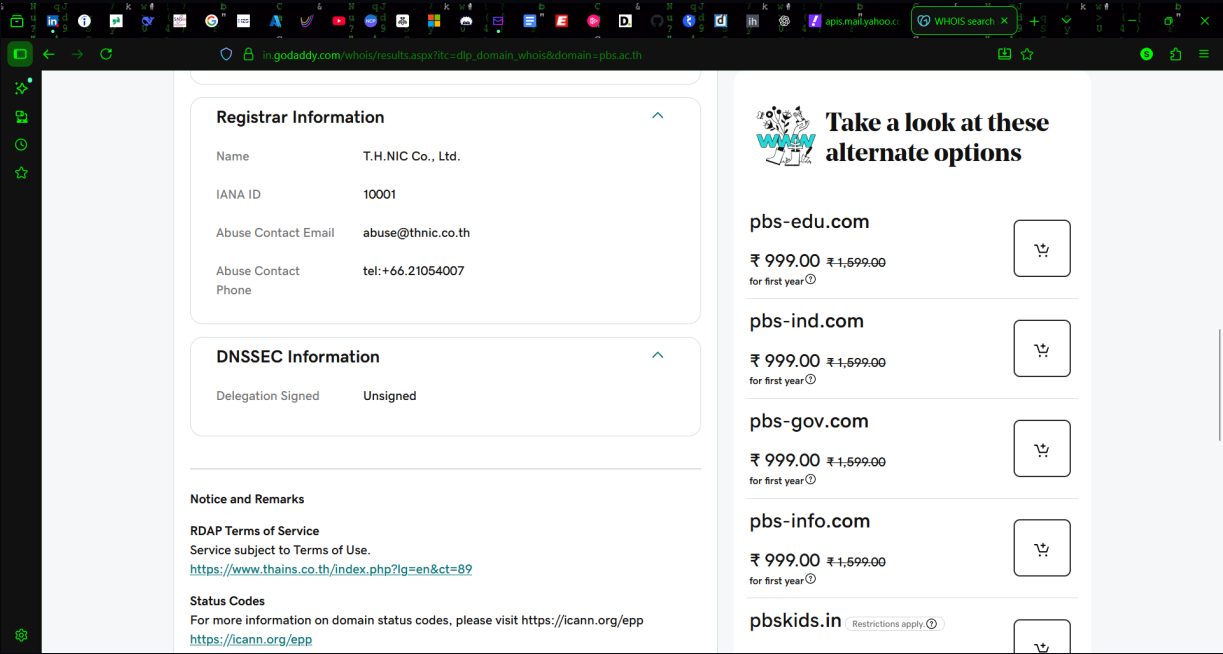


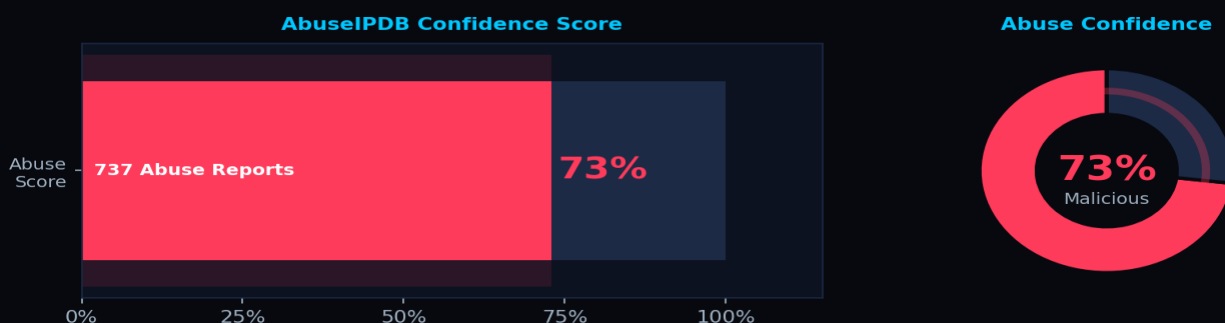
Fig 6. WHOIS registrar information for pbs.ac.th.

06

SOURCE IP ADDRESS ANALYSIS

The originating IP extracted from headers: **209.85.219.66** — Google LLC (AS15169)

IP Address	209.85.219.66
Owner / ISP	Google LLC
ASN	AS15169
Hostname	mail-qv1-f66.google.com
Country	United States of America — Charlotte, NC
Usage Type	Data Center / Web Hosting / Transit



AbuseIPDB: 737 reports · 73% confidence of abuse · Categories: Email Spam, Brute Force, Port Scanning. IP belongs to Google infrastructure — abuse is likely from **compromised Gmail accounts**.

SCREENSHOT — AbuseIPDB Analysis

The screenshot shows the AbuseIPDB website interface. At the top, there's a navigation bar with links like 'Report IP', 'Bulk Checker', 'Bulk Reporter', 'Pricing', 'Docs', 'IP Utilities', 'Contact', and 'More'. A search bar is present with the text 'Check an IP Address, Domain Name, Subnet, or ASN' and a 'CHECK' button. The main content area displays the report for IP 209.85.219.66, stating it was found in the database and has been reported 737 times with a 73% confidence of abuse. A progress bar shows the 73% confidence level. Below this, a table lists details: ISP (Google LLC), Usage Type (Data Center/Web Hosting/Transit), ASN (AS15169), Hostname(s) (mail-qv1-f66.google.com), Domain Name (google.com), Country (United States of America), and City (Charlotte, North Carolina). A footer note mentions that IP info is provided by IPInfo and updated biweekly.

Fig 7. AbuseIPDB report for IP 209.85.219.66.

07

VIRUSTOTAL IP REPUTATION CHECK

Vendors Flagged	0 / 95 — No direct malware association
Suspicious Indicator	1 (CriminalIP flagged as Suspicious)
Infrastructure	Google LLC / AS15169
Conclusion	No malware found, but email abuse behaviour confirmed by AbuseIPDB

SCREENSHOT — VirusTotal IP Reputation Check

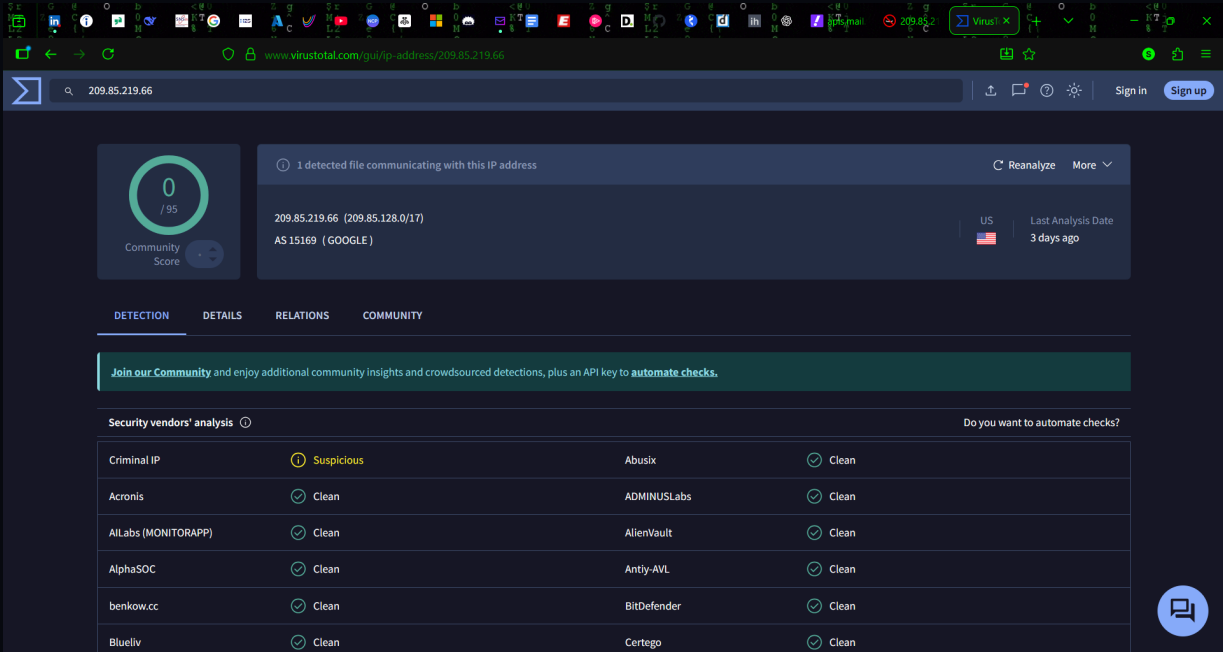


Fig 8. VirusTotal results for IP 209.85.219.66.

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EMAIL CONTENT ANALYSIS

Attachments	None detected
Clickable Links	None detected
Social Engineering	Business supplier lure requesting contract discussion
Reply-To	Differs from sender domain — strong BEC indicator
Threat Type	Business Email Compromise (BEC) / Scam

Threat Assessment: Likely BEC / Scam. Designed to initiate conversation before requesting sensitive or financial data. No malware payload present — purely social engineering based.

SCREENSHOT — Email Body Content



Fig 9. Email body — "Good Morning," social engineering lure.

09

FINAL INCIDENT VERDICT

■ VERDICT: SUSPICIOUS / SCAM EMAIL

Classification: Low-Level Phishing / Business Email Compromise (BEC) Attempt

REASONS FOR CLASSIFICATION

- ✗ Weak email authentication — SPF = none, DMARC = unknown
- ✗ High IP abuse reputation — 73% confidence, 737 reports (AbuseIPDB)
- ✗ Generic business lure content — classic BEC social engineering pattern
- ✗ Email delivered directly to Spam by Yahoo Mail filters
- ✗ Reply-To address mismatch with the sender domain

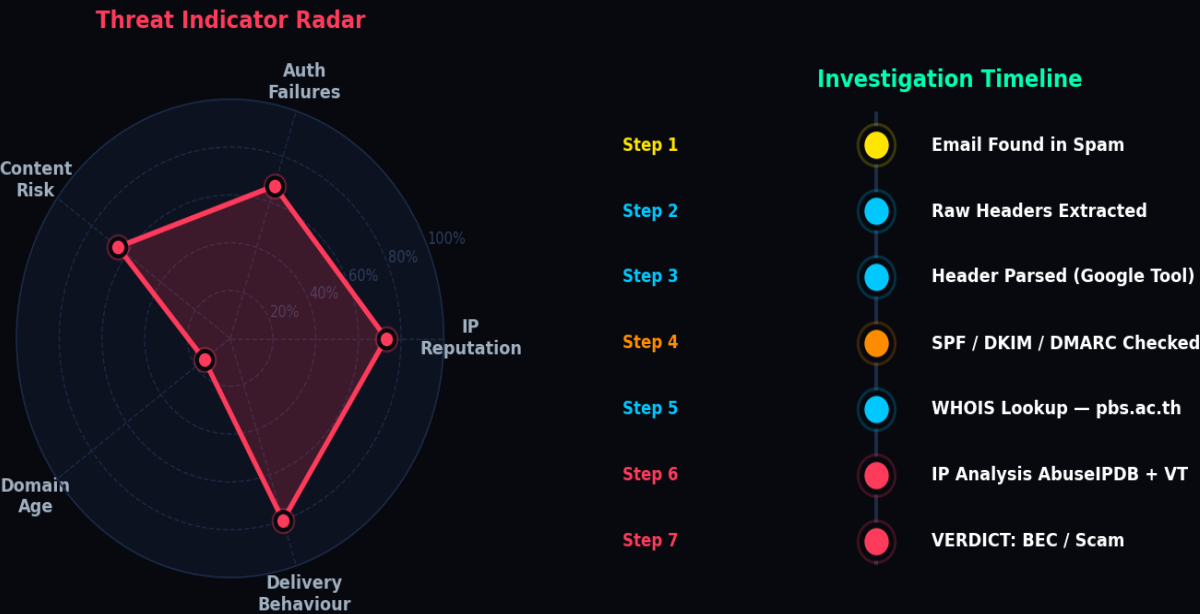


Fig 9. Threat indicator radar (left) and full investigation timeline (right).

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RECOMMENDED SOC ACTIONS

IMMEDIATE	Mark email as Spam — block sender address immediately
IMMEDIATE	Report IP 209.85.219.66 abuse activity on AbuseIPDB
SHORT-TERM	Educate users on BEC-style social engineering tactics
SHORT-TERM	Monitor for similar sender patterns and lookalike domains
LONG-TERM	Enforce SPF and DMARC policies on all organisational domains
LONG-TERM	Deploy email filtering rules for generic business-lure subjects

73% Abuse
Score

737 Abuse
Reports

0/95 VT
Detections

BEC Threat
Type